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### BATTERY STORAGE SYSTEM FOR ELECTRIC REFUSE VEHICLE

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#### Abstract

A refuse vehicle including a chassis, a body assembly coupled to the chassis, the body assembly defining a refuse compartment, and an electric energy system, the electric energy system including one or more battery cells and control hardware, the electric energy system detachably coupled to the body and configured to be accessed by a door in the body, and wherein the one or more battery cells of the electric energy system are replaceable.

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## **Background/Summary**

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS [0001] The present Application is a continuation of U.S. application Ser. No. 16/851,149, filed Apr. 17, 2020, which claims the benefit and priority to U.S. Provisional Patent Application No. 62/842,934, filed on May 3, 2019, the entire disclosure of which is incorporated by reference herein.

### **BACKGROUND**

[0002] Refuse vehicles collect a wide variety of waste, trash, and other material from residences and businesses. Operators of the refuse vehicles transport the material from various waste receptacles within a municipality to a storage or processing facility (e.g., a landfill, an incineration facility, a recycling facility, etc.).

### **SUMMARY**

[0003] One embodiment relates to a refuse vehicle including a chassis, a body assembly coupled to the chassis, the body assembly defining a refuse compartment, and an electric energy system, the electric energy system including one or more battery cells and control hardware, the electric energy system detachably coupled to the body and configured to be accessed by a door in the body, and wherein the one or more battery cells of the electric energy system are replaceable.

[0004] In some embodiments, the refuse vehicle is fully powered by the electric energy system. In some embodiments, the refuse vehicle does not include a combustion engine. In some embodiments, the one or more battery cells are detachably coupled to a first portion of the body and wherein the control hardware is detachably coupled to a second portion of the body distinct from the first portion. In some embodiments, at least a portion of the electric energy system is coupled to a tailgate of the refuse vehicle, and wherein the tailgate is detachably coupled to the body of the refuse vehicle. In some embodiments, at least a portion of the electric energy system is coupled to at least one wheel of the refuse vehicle. In some embodiments, the refuse vehicle includes a cab and wherein at least a portion of the electric energy system is positioned between the cab and the body.

[0005] Another embodiment relates to an electrical energy system for powering a refuse vehicle, the electrical energy system including one or more battery cells configured to store electrical potential energy, control hardware configured to manage charging the one or more battery cells and distributing the stored electrical potential energy, and an attachment assembly configured to detachably couple the electrical energy system to the refuse vehicle, and wherein the electrical energy system is configured to provide power to the refuse vehicle.

[0006] In some embodiments, the electrical energy system is configured to provide an entire power needed to operate the refuse vehicle. In some embodiments, the attachment assembly is configured to detachably couple the electrical energy system to a tailgate of the refuse vehicle. In some embodiments, the attachment assembly is configured to detachably couple the electrical energy system above a cab portion of the refuse vehicle. In some embodiments, the attachment assembly is configured to detachably couple a first component of the electrical energy system to a first portion

of the refuse vehicle and wherein the attachment assembly is configured to detachably couple a second component of the electrical energy system to a second portion of the refuse vehicle that is distinct from the first portion of the refuse vehicle. In some embodiments, the first portion includes a wheel of the refuse vehicle and wherein the second portion includes a body of the electric refuse vehicle. In some embodiments, the attachment assembly is configured to facilitate removal of the one or more battery cells from the refuse vehicle for charging.

[0007] Another embodiment relates to an electric refuse vehicle including a chassis, a body assembly coupled to the chassis, the body assembly defining a refuse compartment, a prime mover, wherein the prime mover is an electric motor, and an electrical energy system including one or more battery cells detachably coupled to the body and configured to completely power the prime mover.

[0008] In some embodiments, a first number of the one or more battery cells are detachably coupled to a first portion of the body and wherein a second number of the one or more battery cells are detachably coupled to a second portion of the body that is distinct from the first portion of the body. In some embodiments, at least one of the one or more battery cells are detachably coupled to a tailgate of the electric refuse vehicle. In some embodiments, the one or more battery cells are hot-swappable. In some embodiments, at least one of the one or more battery cells are positioned between the chassis and the body and are accessible by a door in the body. In some embodiments, the one or more battery cells are positioned between the body and a cab of the electric refuse vehicle.

[0009] In some embodiments, the body assembly is detachably coupled to the chassis. In some embodiments, the electric refuse vehicle is a rear loading refuse vehicle and wherein a center of mass of the electrical energy system is forward of a rear axle of the electric refuse vehicle. In some embodiments, the electric refuse vehicle is a side loading refuse vehicle and wherein a center of mass of the electrical energy system is rearward of a rear axle of the electric refuse vehicle. In some embodiments, the electric refuse vehicle is a front loading refuse vehicle and wherein a center of mass of the electrical energy system is rearward of a rear axle of the electric refuse vehicle.

[0010] This summary is illustrative only and is not intended to be in any way limiting. Other aspects, inventive features, and advantages of the devices or processes described herein will become apparent in the detailed description set forth herein, taken in conjunction with the accompanying figures, wherein like reference numerals refer to like elements.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0011] FIG. 1 is a perspective view of a refuse vehicle, according to an exemplary embodiment.

[0012] FIG. 2 is a side view of a refuse container of the refuse vehicle of FIG. 1 having a top energy storage and/or generation system, according to an exemplary embodiment.

[0013] FIG. 3 is the refuse container of FIG. 1 having a bottom energy storage and/or generation system, according to an exemplary embodiment.

[0014] FIG. 4 is the refuse vehicle of FIG. 1 having a centralized energy storage and/or generation system, according to an exemplary embodiment.

[0015] FIG. 5 is the refuse vehicle of FIG. 1 having a tailgate energy storage and/or generation system, according to an exemplary embodiment.

[0016] FIG. 6 is the refuse vehicle of FIG. 1 having a frame energy storage and/or generation system, according to an exemplary embodiment.

[0017] FIGS. 7A-8B are the refuse vehicle of FIG. 1 having a distributed energy storage and/or generation system, according to several exemplary embodiments.

[0018] FIGS. 9A-9B are the refuse vehicle of FIG. 1 having a top energy storage and/or generation

system, according to several exemplary embodiments.

## DETAILED DESCRIPTION

[0019] Before turning to the figures, which illustrate certain exemplary embodiments in detail, it should be understood that the present disclosure is not limited to the details or methodology set forth in the description or illustrated in the figures. It should also be understood that the terminology used herein is for the purpose of description only and should not be regarded as limiting.

[0020] According to an exemplary embodiment, an energy storage and/or generation system for a refuse vehicle is disclosed herein. The energy storage and/or generation system of the present disclosure provides many advantages over conventional systems. The energy storage and/or generation system may be positioned to evenly distribute the weight of batteries across the frame of the refuse vehicle and/or minimize component stress of one or more load bearing members (e.g., an axle) of the refuse vehicle. The energy storage and/or generation system may be positioned to be easily accessible and/or removable from the refuse vehicle. Ease of access and removability reduce the labor involved in servicing an energy storage and/or generation system, making routine inspection and servicing more feasible and thereby increasing the life of the energy storage and/or generation system. Furthermore, removability allows the energy storage and/or generation system to be “hot-swapped” when it is depleted of charge for a fresh battery, thereby enabling greater uptime for a refuse vehicle. In addition, a removable energy storage and/or generation system may be safely charged at greater speeds than an energy storage and/or generation system confined to a refuse vehicle, thereby allowing for a smaller number of energy storage and/or generation systems to be used to support a fleet of refuse vehicles. Finally, the energy storage and/or generation system may be modular, allowing individual components of the energy storage and/or generation system to be easily replaced for one another. Modularity not only reduces maintenance costs but also allows for future upgrades to the energy storage and/or generation system. For example, the batteries of the energy storage and/or generation system may be easily upgraded to future chemistries not yet available.

### Overall Vehicle

[0021] As shown in FIG. 1, a vehicle, shown as refuse vehicle **10** (e.g., a garbage truck, a waste collection truck, a sanitation truck, a recycling truck, etc.), is configured as a front-loading refuse truck. In other embodiments, the refuse vehicle **10** is configured as a side-loading refuse truck or a rear-loading refuse truck. In still other embodiments, the vehicle is another type of vehicle (e.g., a skid-loader, a telehandler, a plow truck, a boom lift, etc.). As shown in FIG. 1, the refuse vehicle **10** includes a chassis, shown as frame **12**; a body assembly, shown as body **14**, coupled to the frame **12** (e.g., at a rear end thereof, etc.); and a cab, shown as cab **16**, coupled to the frame **12** (e.g., at a front end thereof, etc.). The cab **16** may include various components to facilitate operation of the refuse vehicle **10** by an operator (e.g., a seat, a steering wheel, actuator controls, a user interface, switches, buttons, dials, etc.).

[0022] As shown in FIG. 1, the refuse vehicle **10** includes a prime mover, shown as electric motor **18**, and an energy system, shown as energy storage and/or generation system **20**. In other embodiments, the prime mover is or includes an internal combustion engine. According to the exemplary embodiment shown in FIG. 1, the electric motor **18** is coupled to the frame **12** at a position beneath the cab **16**. The electric motor **18** is configured to provide power to a plurality of tractive elements, shown as wheels **22** (e.g., via a drive shaft, axles, etc.). In other embodiments, the electric motor **18** is otherwise positioned and/or the refuse vehicle **10** includes a plurality of electric motors to facilitate independently driving one or more of the wheels **22**. In still other embodiments, the electric motor **18** or a secondary electric motor is coupled to and configured to drive a hydraulic system that powers hydraulic actuators. According to the exemplary embodiment shown in FIG. 1, the energy storage and/or generation system **20** is coupled to the frame **12** beneath the body **14**. In other embodiments, the energy storage and/or generation system **20** is otherwise

positioned (e.g., within a tailgate of the refuse vehicle **10**, beneath the cab **16**, along the top of the body **14**, within the body **14**, etc.).

[0023] According to an exemplary embodiment, the energy storage and/or generation system **20** is configured to (a) receive, generate, and/or store power and (b) provide electric power to (i) the electric motor **18** to drive the wheels **22**, (ii) electric actuators of the refuse vehicle **10** to facilitate operation thereof (e.g., lift actuators, tailgate actuators, packer actuators, grabber actuators, etc.), and/or (iii) other electrically operated accessories of the refuse vehicle **10** (e.g., displays, lights, etc.). The energy storage and/or generation system **20** may include one or more rechargeable batteries (e.g., lithium-ion batteries, nickel-metal hydride batteries, lithium-ion polymer batteries, lead-acid batteries, nickel-cadmium batteries, etc.), capacitors, solar cells, generators, power buses, etc. In one embodiment, the refuse vehicle **10** is a completely electric refuse vehicle. In other embodiments, the refuse vehicle **10** includes an internal combustion generator that utilizes one or more fuels (e.g., gasoline, diesel, propane, natural gas, hydrogen, etc.) to generate electricity to charge the energy storage and/or generation system **20**, power the electric motor **18**, power the electric actuators, and/or power the other electrically operated accessories (e.g., a hybrid refuse vehicle, etc.). For example, the refuse vehicle **10** may have an internal combustion engine augmented by the electric motor **18** to cooperatively provide power to the wheels **22**. The energy storage and/or generation system **20** may thereby be charged via an on-board generator (e.g., an internal combustion generator, a solar panel system, etc.), from an external power source (e.g., overhead power lines, mains power source through a charging input, etc.), and/or via a power regenerative braking system, and provide power to the electrically operated systems of the refuse vehicle **10**. In some embodiments, the energy storage and/or generation system **20** includes a heat management system (e.g., liquid cooling, heat exchanger, air cooling, etc.).

[0024] According to an exemplary embodiment, the refuse vehicle **10** is configured to transport refuse from various waste receptacles within a municipality to a storage and/or processing facility (e.g., a landfill, an incineration facility, a recycling facility, etc.). As shown in FIG. **1**, the body **14** includes a plurality of panels, shown as panels **32**, a tailgate **34**, and a cover **36**. The panels **32**, the tailgate **34**, and the cover **36** define a collection chamber (e.g., hopper, etc.), shown as refuse compartment **30**. Loose refuse may be placed into the refuse compartment **30** where it may thereafter be compacted (e.g., by a packer system, etc.). The refuse compartment **30** may provide temporary storage for refuse during transport to a waste disposal site and/or a recycling facility. In some embodiments, at least a portion of the body **14** and the refuse compartment **30** extend above or in front of the cab **16**. According to the embodiment shown in FIG. **1**, the body **14** and the refuse compartment **30** are positioned behind the cab **16**. In some embodiments, the refuse compartment **30** includes a hopper volume and a storage volume. Refuse may be initially loaded into the hopper volume and thereafter compacted into the storage volume. According to an exemplary embodiment, the hopper volume is positioned between the storage volume and the cab **16** (e.g., refuse is loaded into a position of the refuse compartment **30** behind the cab **16** and stored in a position further toward the rear of the refuse compartment **30**, a front-loading refuse vehicle, a side-loading refuse vehicle, etc.). In other embodiments, the storage volume is positioned between the hopper volume and the cab **16** (e.g., a rear-loading refuse vehicle, etc.).

[0025] As shown in FIG. **1**, the refuse vehicle **10** includes a lift mechanism/system (e.g., a front-loading lift assembly, etc.), shown as lift assembly **40**, coupled to the front end of the body **14**. In other embodiments, the lift assembly **40** extends rearward of the body **14** (e.g., a rear-loading refuse vehicle, etc.). In still other embodiments, the lift assembly **40** extends from a side of the body **14** (e.g., a side-loading refuse vehicle, etc.). As shown in FIG. **1**, the lift assembly **40** is configured to engage a container (e.g., a residential trash receptacle, a commercial trash receptacle, a container having a robotic grabber arm, etc.), shown as refuse container **60**. The lift assembly **40** may include various actuators (e.g., electric actuators, hydraulic actuators, pneumatic actuators, etc.) to facilitate engaging the refuse container **60**, lifting the refuse container **60**, and tipping refuse

out of the refuse container **60** into the hopper volume of the refuse compartment **30** through an opening in the cover **36** or through the tailgate **34**. The lift assembly **40** may thereafter return the empty refuse container **60** to the ground. According to an exemplary embodiment, a door, shown as top door **38**, is movably coupled along the cover **36** to seal the opening thereby preventing refuse from escaping the refuse compartment **30** (e.g., due to wind, bumps in the road, etc.).

#### Energy Storage and/or Generation System

[0026] As shown in FIG. 2, the energy storage and/or generation system **20** is coupled to the rearward top portion of the body **14**. In other embodiments, the energy storage and/or generation system **20** is coupled to the forward top portion of the body **14**. In some embodiments, the energy storage and/or generation system **20** is removable/detachable from the body **14**. Locating the energy storage and/or generation system **20** on top of the body **14** facilitates easy access to the energy storage and/or generation system **20**. For example, a user may readily inspect and service the energy storage and/or generation system **20** because it is located on an external surface of the refuse vehicle **10**.

[0027] The energy storage and/or generation system **20** may include one or more attachment mechanisms (e.g., pins, linkages, latches, etc.) to couple the energy storage and/or generation system **20** to the body **14**. In some embodiments, the energy storage and/or generation system **20** is a pod or battery compartment, configured to receive and electrically couple one or more batteries. For example, the energy storage and/or generation system **20** may allow a battery cell to be inserted from one end thereby coupling the battery cell to the energy storage and/or generation system **20** and providing power to the refuse vehicle **10**. In some embodiments, the energy storage and/or generation system **20** is modular and facilitates easy replacement of one or more battery cells. For example, a second fully charged battery cell may replace a first depleted battery cell by uncoupling the first battery cell from the energy storage and/or generation system **20** and replacing it with the second battery cell. In some embodiments, the entire energy storage and/or generation system **20** is replaced with a different one of energy storage and/or generation system **20**. Replacing one or more battery cells of the energy storage and/or generation system **20** reduces the downtime associated with charging a typical battery system. In some embodiments, the energy storage and/or generation system **20** is “hot-swappable” and is able to replace one or more battery cells without cutting power to the refuse vehicle **10**.

[0028] The energy storage and/or generation system **20** may include an electric connection (e.g., a pantograph, a current collector, a high-voltage line, etc.) to allow the energy storage and/or generation system **20** to connect to external power sources (e.g., an overhead power line, the grid, a charging station, etc.). For example, the energy storage and/or generation system **20** may include a charging port to allow one or more battery cells to be charged while the energy storage and/or generation system **20** is coupled to the refuse vehicle **10** (e.g., by a 220V charger). In some embodiments, the energy storage and/or generation system **20** includes an electrical bypass to power the refuse vehicle **10** from a charging source while the battery is being charged. In some embodiments, the energy storage and/or generation system **20** connects to one or more power sources of refuse vehicle **10** (e.g., an internal combustion generator, a battery, etc.) to charge the energy storage and/or generation system **20**. For example, the energy storage and/or generation system **20** may include a connection to an onboard diesel generator configured to provide power to the energy storage and/or generation system **20** for charging.

[0029] As shown in FIG. 3, the energy storage and/or generation system **20** is coupled to the rearward bottom portion of the body **14**. In other embodiments, the energy storage and/or generation system **20** is coupled to the forward bottom portion of the body **14**. As described above, the energy storage and/or generation system **20** may be removable/replaceable. For example, the refuse vehicle **10** may include a door on the side of the body **14** to allow removal and replacement of the energy storage and/or generation system **20**. In some embodiments, the energy storage and/or generation system **20** is located on a track such that the energy storage and/or generation system **20**

can slide out from the body **14** similar to a drawer. In some embodiments, the energy storage and/or generation system **20** is modular. For example, the energy storage and/or generation system **20** may include one or more sub-batteries to reduce the bulkiness of the energy storage and/or generation system **20** and permit easy removal and/or replacement. Modularity further enables more precise inspection and service of battery cells and allows individual battery cells to be replaced without the need to replace an entire larger array. Furthermore, modularity enables battery cells to be easily upgraded.

[0030] As described above, the energy storage and/or generation system **20** may include a charging port to allow the energy storage and/or generation system **20** to receive external power for charging. For example, the refuse vehicle **10** may include a 220V charging port on a side of the body **14** to charge the energy storage and/or generation system **20**.

[0031] As shown in FIG. **4**, the energy storage and/or generation system **20** is coupled between the cab **16** and the body **14**. In some embodiments, the energy storage and/or generation system **20** is coupled to the frame **12**. Locating the energy storage and/or generation system **20** between the cab **16** and the body **14** reduces a rear weight of the refuse vehicle **10**, thereby reducing component stress of weight bearing members (e.g., a rear axle). Furthermore, centrally locating the energy storage and/or generation system **20** protects the energy storage and/or generation system **20** from damage in the event of a collision. Furthermore, centrally locating the energy storage and/or generation system **20** allows easy modification/retrofitting of existing refuse vehicles to include the energy storage and/or generation system **20**. The energy storage and/or generation system **20** may be easily accessed and/or removed from the refuse vehicle **10**. For example, the energy storage and/or generation system **20** may include forklift pockets so that a forklift may easily remove the energy storage and/or generation system **20** from the refuse vehicle **10**. In some embodiments, the system **20** includes one or more eyelet connectors to receive a lifting hook or similar hoisting attachment. The energy storage and/or generation system **20** may be configured to connect to an external rail system to quickly replace the energy storage and/or generation system **20** by sliding it orthogonally off the refuse vehicle **10**.

[0032] In some embodiments, the energy storage and/or generation system **20** is configured to dynamically change position on the refuse vehicle **10** based on loading of the refuse vehicle **10**. For example, the energy storage and/or generation system **20** may translate horizontally along the frame **12** toward the cab **16** or toward the body **14** to change a weight distribution of the vehicle. In some embodiments, the energy storage and/or generation system **20** includes one or more controllers to measure the weight distribution of the refuse vehicle **10** and adjust a position of the energy storage and/or generation system **20** accordingly.

[0033] As shown in FIG. **5**, the energy storage and/or generation system **20** is coupled to the tailgate **34** of the refuse vehicle **10**. In some embodiments, the energy storage and/or generation system **20** is positioned vertically along a rearward side of the refuse compartment **30**. In some embodiments, the energy storage and/or generation system **20** is positioned substantially near the base of the tailgate **34** or as part of the tailgate **34**. The energy storage and/or generation system **20** may be configured to be accessible via the tailgate **34**. For example, a user could open the tailgate **34** to reveal the energy storage and/or generation system **20**. In some embodiments, the tailgate **34** includes one or more rotating elements (e.g., hinges, mechanical bearings) to facilitate rotation around a rearward corner of the refuse compartment **30**. For example, the tailgate **34** could include one or more hinging mechanisms on a side to allow a user to open the tailgate **34** like a door and gain access to the energy storage and/or generation system **20** located along the frame **12** of the refuse vehicle **10**. In some embodiments, the tailgate **34** is a double door. Swinging the tailgate **34** open like a door requires less energy than lifting the tailgate **34**.

[0034] In some embodiments, the tailgate **34** is fully integrated with the energy storage and/or generation system **20** and is configured to be removable/replaceable. For example, a first tailgate **34** having a first energy storage and/or generation system **20** could be replaced by a second tailgate

**34** having a second energy storage and/or generation system **20** when the first energy storage and/or generation system **20** is depleted of energy. Removing and replacing the tailgate **34** may limit loss of vehicle operation due to charging time because the tailgate **34** including the depleted energy storage and/or generation system **20** may be charged separately of the refuse vehicle **10**. Furthermore, swappable energy storage and/or generation systems enable a smaller fleet of refuse vehicles to service the same area because the reduced downtime associated with battery charging enables the refuse vehicles to operate for longer periods of time. In some embodiments, a number of racks index one or more battery cells of the energy storage and/or generation system **20**.

[0035] As shown in FIG. 6, the energy storage and/or generation system **20** is coupled between the body **14** and the frame **12**. As described above, in some embodiments, the energy storage and/or generation system **20** may be configured to translate horizontally along the frame **12** of the refuse vehicle **10**. For example, the energy storage and/or generation system **20** could move between a forward portion and a rearward portion of the body **14** of the refuse vehicle **10** such that the refuse vehicle **10** is evenly loaded. As described above, in some embodiments, the energy storage and/or generation system **20** is removable and/or replaceable. The energy storage and/or generation system **20** may be accessed via a door on a side of the body **14** or via the tailgate **34**. Similarly, the energy storage and/or generation system **20** may be removed and/or replaced by another energy storage and/or generation system. Alternatively, one or more individual battery cells of the energy storage and/or generation system **20** could be replaced. In some embodiments, the energy storage and/or generation system **20** can be accessed by removing the refuse compartment **30**. For example, a refuse vehicle with a removable refuse compartment (e.g., a container truck) may remove the refuse compartment to reveal the energy storage and/or generation system **20**. In some embodiments, the energy storage and/or generation system **20** is coupled to the refuse compartment **30** itself and can be removed with the refuse compartment **30**. For example, a refuse vehicle could swap a first full refuse compartment with a first depleted energy storage and/or generation system for a second empty refuse compartment with a second charged energy storage and/or generation system.

[0036] Referring now to FIGS. 7A-8B, several illustrations of an exemplary placement of the energy storage and/or generation system **20** is shown, according to several exemplary embodiments. In various embodiments, the energy storage and/or generation system **20** is coupled to a rearward top portion of the refuse vehicle **10** (e.g., above the refuse compartment **30**, etc.). Additionally or alternatively, the energy storage and/or generation system **20** is coupled to a rearward portion of the refuse vehicle **10**. For example, the energy storage and/or generation system **20** may be coupled to the tailgate **34** and/or a rearward portion of the refuse compartment **30** (e.g., as shown in FIGS. 7A-7C). As another example, the energy storage and/or generation system **20** may be coupled to a vertical rear surface of the refuse compartment **30**. In some embodiments, the energy storage and/or generation system **20** or components thereof are coupled to the wheel **22**. For example, an energy storage cell of the energy storage and/or generation system **20** may be coupled to a hub of the wheels **22** and a power converter of the energy storage and/or generation system **20** may be coupled to a top rearward portion of the refuse container **30**. In some embodiments, the energy storage and/or generation system **20** is coupled to a front and rear wheelset of the refuse vehicle **10** (e.g., as shown in FIGS. 7A-7C). In various embodiments, placement of the energy storage and/or generation system **20** as shown in FIGS. 7A-7C facilitates shifting weight rearward on the refuse vehicle **10**, thereby reducing strain on forward load bearing components (e.g., a front axle, etc.). In some embodiments, the placement of the energy storage and/or generation system **20** shown in FIGS. 7A-7C is preferred for a rear-loading refuse vehicle **10**. In various embodiments, the energy storage and/or generation system **20** includes a different number and/or arrangement of components than shown explicitly in the FIGURES. For example, the energy storage and/or generation system **20** may include a first component coupled to an exterior hub surface of the front wheels **22** electrically coupled to a second component integrated



with the tailgate **34**. In some embodiments, the placement of the energy storage and/or generation system **20** shown in FIGS. **8A-8B** is preferred for a front-loading refuse vehicle **10** and/or a side-loading refuse vehicle **10**. In various embodiments, the energy storage and/or generation system **20**, or components thereof, are detachable from the refuse vehicle **10** as described in detail above. [0037] Referring now to FIGS. **9A-9B**, several illustrations of another exemplary placement of the energy storage and/or generation system **20** is shown, according to several exemplary embodiments. In various embodiments, the energy storage and/or generation system **20** is coupled to a top portion of the refuse vehicle **10**. For example, the energy storage and/or generation system **20** may be coupled to a top portion of refuse compartment **30** and/or above the cab **16** (e.g., as shown in FIGS. **9A-9B**). In some embodiments, the energy storage and/or generation system **20** is coupled to a canopy (or other structural element) located above the cab **16**. Additionally or alternatively, the energy storage and/or generation system **20**, or components thereof, may be coupled to the wheels **22**. For example, a first component of the energy storage and/or generation system **20** (e.g., a battery cell, etc.) may be coupled to an exterior hub region of the wheels **22** and a second component of the energy storage and/or generation system **20** (e.g., a power converter, etc.) may be coupled to a structural element (e.g., a portion of frame **12**, etc.) above the cab **16**. In some embodiments, the placement of the energy storage and/or generation system **20** shown in FIGS. **9A-9B** is preferred for a rear-loading refuse vehicle **10**. In various embodiments, the placement of the energy storage and/or generation system **20** as shown in FIGS. **9A-9B** facilitates moving weight (e.g., battery weight, etc.) forward on the refuse vehicle **10** (e.g., toward the cab **16** and away from the tailgate **34**, etc.), thereby reducing stress on rear load-bearing components (e.g., a rear axle, etc.).

[0038] As utilized herein, the terms “approximately,” “about,” “substantially”, and similar terms are intended to have a broad meaning in harmony with the common and accepted usage by those of ordinary skill in the art to which the subject matter of this disclosure pertains. It should be understood by those of skill in the art who review this disclosure that these terms are intended to allow a description of certain features described and claimed without restricting the scope of these features to the precise numerical ranges provided. Accordingly, these terms should be interpreted as indicating that insubstantial or inconsequential modifications or alterations of the subject matter described and claimed are considered to be within the scope of the disclosure as recited in the appended claims.

[0039] It should be noted that the term “exemplary” and variations thereof, as used herein to describe various embodiments, are intended to indicate that such embodiments are possible examples, representations, or illustrations of possible embodiments (and such terms are not intended to connote that such embodiments are necessarily extraordinary or superlative examples).

[0040] The term “coupled” and variations thereof, as used herein, means the joining of two members directly or indirectly to one another. Such joining may be stationary (e.g., permanent or fixed) or moveable (e.g., removable or releasable). Such joining may be achieved with the two members coupled directly to each other, with the two members coupled to each other using a separate intervening member and any additional intermediate members coupled with one another, or with the two members coupled to each other using an intervening member that is integrally formed as a single unitary body with one of the two members. If “coupled” or variations thereof are modified by an additional term (e.g., directly coupled), the generic definition of “coupled” provided above is modified by the plain language meaning of the additional term (e.g., “directly coupled” means the joining of two members without any separate intervening member), resulting in a narrower definition than the generic definition of “coupled” provided above. Such coupling may be mechanical, electrical, or fluidic.

[0041] References herein to the positions of elements (e.g., “top,” “bottom,” “above,” “below”) are merely used to describe the orientation of various elements in the FIGURES. It should be noted that the orientation of various elements may differ according to other exemplary embodiments, and

that such variations are intended to be encompassed by the present disclosure.

[0042] The hardware and data processing components used to implement the various processes, operations, illustrative logics, logical blocks, modules and circuits described in connection with the embodiments disclosed herein may be implemented or performed with a general purpose single- or multi-chip processor, a digital signal processor (DSP), an application specific integrated circuit (ASIC), a field programmable gate array (FPGA), or other programmable logic device, discrete gate or transistor logic, discrete hardware components, or any combination thereof designed to perform the functions described herein. A general purpose processor may be a microprocessor, or, any conventional processor, controller, microcontroller, or state machine. A processor also may be implemented as a combination of computing devices, such as a combination of a DSP and a microprocessor, a plurality of microprocessors, one or more microprocessors in conjunction with a DSP core, or any other such configuration. In some embodiments, particular processes and methods may be performed by circuitry that is specific to a given function. The memory (e.g., memory, memory unit, storage device) may include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present disclosure. The memory may be or include volatile memory or non-volatile memory, and may include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present disclosure. According to an exemplary embodiment, the memory is communicably connected to the processor via a processing circuit and includes computer code for executing (e.g., by the processing circuit or the processor) the one or more processes described herein.

[0043] The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

[0044] Although the figures and description may illustrate a specific order of method steps, the order of such steps may differ from what is depicted and described, unless specified differently above. Also, two or more steps may be performed concurrently or with partial concurrence, unless specified differently above. Such variation may depend, for example, on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations of the described methods could be accomplished with standard programming techniques with rule-based logic and other logic to accomplish the various connection steps, processing steps, comparison steps, and decision steps.

[0045] It is important to note that the construction and arrangement of the refuse vehicle **10** and the systems and components thereof as shown in the various exemplary embodiments is illustrative only. Additionally, any element disclosed in one embodiment may be incorporated or utilized with

any other embodiment disclosed herein. Although only one example of an element from one embodiment that can be incorporated or utilized in another embodiment has been described above, it should be appreciated that other elements of the various embodiments may be incorporated or utilized with any of the other embodiments disclosed herein.

## Claims

1. A refuse vehicle, comprising: a chassis; a body coupled to the chassis, the body defining a refuse compartment including a hopper portion and a storage portion, and wherein the body includes a tailgate; and an electrical energy system, the electrical energy system comprising a plurality of battery cells positioned within a modular housing, the electrical energy system detachably coupled to the body and accessible through the tailgate, the tailgate configured to be opened such that the plurality of battery cells are removable and replaceable through the tailgate.
2. The refuse vehicle of claim 1, wherein the refuse vehicle is fully powered by the electrical energy system.
3. The refuse vehicle of claim 1, wherein the refuse vehicle does not include a combustion engine.
4. The refuse vehicle of claim 1, wherein the refuse vehicle includes a cab and wherein at least a portion of the electrical energy system is positioned between the cab and the body.
5. The refuse vehicle of claim 1, wherein the body is a first body removably coupled to the chassis and the electrical energy system is a first electrical energy system, the first body configured to be removed from the chassis with the first electrical energy system coupled to the first body and replaced with a second body having a second electrical energy system at a full charge.
6. The refuse vehicle of claim 1, wherein the electrical energy system extends between the body and the chassis.
7. The refuse vehicle of claim 1, wherein the electrical energy system is selectively repositionable relative to the chassis.
8. The refuse vehicle of claim 1, further comprising a cab coupled to the chassis, wherein the hopper portion of the refuse compartment extends between the storage portion and the cab.
9. The refuse vehicle of claim 1, wherein at least one of the plurality of battery cells extends below a top surface of the chassis, directly below the storage portion of the refuse compartment, and directly alongside a pair of rear tractive elements of the refuse vehicle in a longitudinal direction.
10. The refuse vehicle of claim 1, wherein the plurality of battery cells of the electrical energy system are configured to provide electrical energy for performing a driving operation of the refuse vehicle.
11. An electric refuse vehicle, comprising: a chassis; a body coupled to the chassis, the body defining a refuse compartment including a hopper portion and a storage portion, and wherein the body includes a tailgate; a prime mover, wherein the prime mover is an electric motor; and an electrical energy system including a plurality of battery cells positioned, wherein the electrical energy system is accessible through the tailgate, wherein the tailgate is configured to be opened such that the plurality of battery cells are removable and replaceable through the tailgate; wherein the plurality of battery cells are configured to provide electrical energy to power the prime mover and the plurality of battery cells are individually removable and replaceable.
12. The electric refuse vehicle of claim 11, wherein the body is detachably coupled to the chassis.
13. The electric refuse vehicle of claim 12, wherein the electric refuse vehicle is a rear loading refuse vehicle and wherein a center of mass of the electrical energy system is forward of a rear axle of the electric refuse vehicle.
14. The electric refuse vehicle of claim 11, wherein the electric refuse vehicle is a side loading refuse vehicle and wherein a center of mass of the electrical energy system is rearward of a rear axle of the electric refuse vehicle.
15. The electric refuse vehicle of claim 11, wherein the electric refuse vehicle is a front loading

refuse vehicle and wherein a center of mass of the electrical energy system is rearward of a rear axle of the electric refuse vehicle.

**16.** The electric refuse vehicle of claim 11, further comprising a cab coupled to the chassis, wherein the hopper portion of the refuse compartment extends between the storage portion and the cab.

**17.** A refuse vehicle, comprising: a frame; a body coupled to the frame, the body defining a refuse compartment, having a bottommost surface, and including a tailgate; and an electric energy system, the electric energy system including a plurality of battery cells positioned within a housing; wherein the electric energy system is accessible through the tailgate; wherein the tailgate is configured to be opened such that the plurality of battery cells are removable and replaceable through the tailgate; wherein the plurality of battery cells of the electric energy system are configured to provide electrical energy for performing a driving operation of the refuse vehicle.

**18.** The refuse vehicle of claim 17, wherein the refuse vehicle is fully powered by the electric energy system.

**19.** The refuse vehicle of claim 17, wherein the refuse vehicle does not include a combustion engine.

**20.** The refuse vehicle of claim 17, wherein the refuse vehicle includes a cab and wherein at least a portion of the electric energy system is positioned between the cab and the body.

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